

Rajendra Prasad Kudumula

Senior Backend Engineer | Full Stack Engineer | AI Integration | Node.js | TypeScript | NestJS
✉ raj kudumala81@gmail.com | ☎ [8328375393](tel:8328375393) | 📍 Bangalore | 🌐 [Git Hub](#) | 🔗 [LinkedIn](#)

Professional Summary

Senior Backend Engineer with **4+ years of experience** building scalable backend and full-stack applications using **Node.js, TypeScript, NestJS, PostgreSQL, MongoDB, and AWS**. Strong experience delivering **RESTful APIs, microservices, database-driven applications, API integrations, and performance-focused backend solutions** with emphasis on scalability, reliability, and maintainability. Hands-on exposure to **Angular and React** for full-stack application integration and frontend-backend workflows. Experienced in building **AI-enabled applications** using **LLM integrations, RAG architectures, vector databases, and AI-assisted development workflows**.

Technical Skills

</> Backend Engineering Node.js, TypeScript, JavaScript (ES6+), NestJS, Express.js, REST API Development, Backend Architecture, gRPC, multer, upload	🗄 Database Engineering PostgreSQL, SQL, Database Design, Query Optimization, Indexing, Joins, Transactions, Data Modeling, Redis
🏗 Software Architecture & System Design Microservices Architecture, Scalable Systems, System Design, API Design, Design Patterns, Authentication, Authorization, Distributed Systems Concepts, Event-driven Architecture, Caching Strategies	🛠 DevOps & Development Tools AWS S3, Docker, Git, GitHub, Postman, VS Code, DBever, CI/CD Concepts, GitLab, JIRA, Slack
🔗 API Quality & Automation Swagger, OpenAPI, Jest, AJV Validation, Unit Testing, API Documentation, Code Reviews, Debugging, Clean Code Practices, Rate Limiting, MS Office, MS Docs, MS PPT, MS Excel Sheet, MS Reports	🧠 AI Engineering & Intelligent Applications Generative AI, LLM Integration, Prompt Engineering, RAG Concepts, Vector Databases, Embeddings, Semantic Search, AI Automation

Professional Experience

<u>Happiest Minds Technologies Limited</u> Senior Software Engineer Software Engineer	Bangalore Jul 2025 To Present Feb 2022 – Jun 2025
<u>PRove IT Catalysts</u> Junior Product Developer (Internship)	Hyderabad Jan 2021 To Feb 2022

Projects

WebMD Ignite (formerly Mercury Healthcare) Tech Stack: Node.js, NestJS, Express.js, TypeScript, JavaScript, Angular 21, React, PostgreSQL, Aurora, Snowflake, AWS S3, Microservices, REST APIs, AJV, Swagger, OpenAPI, Highcharts, ExcelJS, PDFMake, Jest, YAML, Cron Jobs, GitHub Copilot, Claude Code	Aug 2023 To Present
• Owned end-to-end backend and full-stack development across healthcare and campaign-management workflows using Node.js, NestJS, Express.js, TypeScript, and Angular 21, delivering scalable microservices, REST APIs, admin panels, and data-driven application features. • 20+ RESTful APIs optimized across backend workflows and PostgreSQL operations through indexing strategies, query optimization, and efficient data-access patterns, strengthening application performance and response reliability. • Built Angular 21 admin panels and Highcharts-based analytics dashboards, enabling interactive visualization and reporting of operational and campaign data; generated downloadable PDF and Excel reports using PDFMake and ExcelJS. • Developed campaign forecasting and performance workflows integrating 2 external advertising platforms – Google and Meta – to surface key metrics including impressions, CTR, and CPC, while building audience CRUD workflows to associate audiences with campaigns and track campaign-level data. • Integrated Snowflake data workflows and scheduled background processing using SQL queries, Cron Jobs, YAML configurations, and microservice integrations, supporting automated data processing and reporting workflows. • Built reusable internal packages and shared backend components while integrating third-party services, transforming request/response payloads, and maintaining API contracts through AJV, Swagger, and OpenAPI. • Full-stack API integration supported across Angular and React-based applications, collaborating on REST API contracts, request/response structures, validation requirements, and backend-driven application workflows. • AI-assisted development workflows leveraged with ChatGPT, GitHub Copilot, and Claude Opus Claude Code for code analysis, debugging, test-case generation, API documentation, and edge-case validation, contributing to an estimated ~30% reduction in feature turnaround time.	

Marketron | Software Engineer

Apr 2022 – Jun 2023

Tech Stack: Node.js, Express.js, React, MongoDB, AWS, RabbitMQ, REST APIs, Microservices, JWT, Jest, GitHub, CI/CD, PDFMake, PptxGenJS, ExcelJS

- Owned backend development across new microservices and REST APIs using Node.js, Express.js, MongoDB, AWS, and RabbitMQ, supporting proposal, campaign, mapping, and order-management workflows.
- Built automated proposal-generation workflows producing PDF, PowerPoint, and Excel reports using PDFMake, PptxGenJS, and ExcelJS, including dynamic business data and map-based SVG visualizations within generated reports.
- Integrated multiple third-party platforms, including TTD and MatchCraft, transforming internal payloads into partner-specific request formats, processing external responses, and maintaining reliable API-to-API communication.
- Streamlined proposal-to-order and campaign workflows, connecting backend services and business processes to convert proposals into orders while supporting campaign creation and downstream operations.
- Strengthened application performance and service reliability through backend optimization, asynchronous processing with RabbitMQ, API improvements, and efficient MongoDB data-access patterns.
- Implemented JWT-based authentication and automated Jest testing while supporting GitHub-based CI/CD pipelines, production debugging, defect resolution, and continuous service enhancements.

Cukzz Food-Tech Platform

Jul 2021 To Feb 2022

Tech Stack: Node.js, TypeScript, PostgreSQL, MongoDB, AWS S3, Jest, AJV Validation

- Delivered backend features and REST APIs for a food-tech platform connecting customers with home-cooked meal providers, using Node.js, TypeScript, and JavaScript across core application workflows.
- Handled PostgreSQL and MongoDB data workflows including database operations, data management, and backend integration supporting application features and evolving business requirements.
- Strengthened API reliability through AJV schema-based request validation and integrated AWS S3 for file upload and storage workflows across application features.
- Built and maintained Jest unit tests while diagnosing backend issues, resolving feature-level defects, and improving application reliability through structured development and debugging practices.
- Collaborated with senior engineers and product teams on feature requirements, API behavior, debugging, and incremental backend enhancements, gaining hands-on experience with Agile software development and production-oriented engineering practices.

Awards	
Distinct Contributor Award	Q3,FY25
Happiest Minds Technologies Limited	
Team Excellence Award	Q4,FY26
Happiest Minds Technologies Limited	

Achievements & Innovation

Blitz25

Internal Web Application: Independently built and delivered an internal web application using Node.js, NestJS, and ReactJs as part of the company's annual Blitz initiative, demonstrating end-to-end application development and solution ownership.

Education	
S V University	Masters of Computer Science • 7.9
Computer Science	Andhra Pradesh • Jun 2018 - Mar 2020
V S University	Bachelor Degree • 9.2
Electronics & Computer Science	Andhra Pradesh • Jun 2015 - Mar 2018